

INFO8006 Introduction to Artificial Intelligence

Exercise session 3: Quantifying uncertainty

RECAP

Definitions

A probability is a degree of belief in a proposition.

Sample space	Ω , of possible worlds ω
Axioms	$0 \leq P(\omega) \leq 1$, $\sum_{\omega \in \Omega} P(\omega) = 1$
Event	$A \subseteq \Omega$, $P(A) = \sum_{\omega \in A} P(\omega)$
Random variable	$X : \Omega \rightarrow D_X$, with $X = x$ the event $\{\omega \in \Omega : X(\omega) = x\}$
Distribution	$\mathbf{P}(X)$, the numbers $P(x) = P(X = x)$ for $x \in D_X$
Joint distribution	$\mathbf{P}(X_1, \dots, X_n)$, the numbers $P(x_1, \dots, x_n)$
Conditional probability	$P(a \mid b) = P(a, b) / P(b)$, for $P(b) > 0$
Conditional distribution	$\mathbf{P}(X \mid Y)$, one $\mathbf{P}(X \mid y)$ for each $y \in D_Y$

Random variables are written in upper case and their values in lower case; a boolean variable A takes the values a and $\neg a$. $P(x)$ is a number, $\mathbf{P}(X)$ a table.

Rules

Complement	$P(\neg a) = 1 - P(a)$
Union	$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$
Product rule	$P(a, b) = P(a \mid b)P(b) = P(b \mid a)P(a)$
Sum rule, total probability	$P(x) = \sum_y P(x, y) = \sum_y P(x \mid y)P(y)$
Chain rule	$P(x_{1:n}) = \prod_i P(x_i \mid x_{1:i-1})$
Bayes' rule	$P(a \mid b) = P(b \mid a)P(a) / P(b)$
Independence, $X \perp Y$	$P(x, y) = P(x)P(y)$
Conditional independence, $X \perp Y \mid Z$	$P(x, y \mid z) = P(x \mid z)P(y \mid z)$
Naive Bayes	$P(c, e_{1:n}) = P(c) \prod_i P(e_i \mid c)$

The rules hold for all values, and between distributions. Naive Bayes assumes the effects $e_{1:n}$ conditionally independent given the cause c . The joint distribution answers every query: select the entries consistent with the evidence, sum out the other variables, normalize.

Bayes' rule

A model is given in the *causal* direction, a *prior* $\mathbf{P}(A)$ and a *likelihood* $\mathbf{P}(B \mid A)$. The *diagnostic* question is answered by the *posterior* $\mathbf{P}(A \mid b) = \frac{1}{Z} \mathbf{P}(b \mid A) \mathbf{P}(A)$, normalized by the probability of the *evidence*, $Z = P(b) = \sum_a P(b \mid a)P(a)$.

In session: exercises 1, 2, 3, 5.

Exercise 1 Rational beliefs (AIMA ch. 12)

Let $A, B \subseteq \Omega$ be two *events*. An agent holds the beliefs $P(A) = 0.4$ and $P(B) = 0.3$. What ranges of probabilities would it be rational for the agent to hold for the events $A \cup B$ and $A \cap B$? What if the agent also believes that $P(A \mid B) = 0.5$?

Exercise 2 Full joint distribution (AIMA ch. 12)

A monitoring program supervises the bearings of a wind turbine. A worn bearing (*fault*) tends to make the turbine vibrate (*vibration*) and to trigger the alarm of a diagnostic probe (*alarm*). The program holds the joint distribution $\mathbf{P}(V, A, F)$ below, over the boolean random variables V (vibration), A (alarm) and F (fault). Compute the following probabilities.

	v		$\neg v$	
	a	$\neg a$	a	$\neg a$
f	0.108	0.012	0.072	0.008
$\neg f$	0.016	0.064	0.144	0.576

- (a) $P(v)$
- (b) $P(f)$
- (c) $P(v \mid f)$
- (d) $P(f \mid v \vee a)$
- (e) Are V and A independent? Are they conditionally independent given F ?

Exercise 3 Fraud detection (AIMA ch. 12)

A fraud detection program screens card payments. It is 99% accurate, *i.e.* it flags a fraudulent payment with probability 0.99 and a legitimate payment with probability 0.01. Fraud is rare: it strikes only 1 payment in 10 000. The program has just flagged your payment. Why is it good news, for you, that fraud is rare? What is the probability that the payment is actually fraudulent?

Exercise 4 Conditional independence (AIMA ch. 12)

For each of the following statements, either prove it is valid or give a counterexample.

- (a) If $P(a, b, c) > 0$ and $P(a \mid b, c) = P(b \mid a, c)$, then $P(a \mid c) = P(b \mid c)$.
- (b) If $P(a \mid b, c) = P(a)$, then $P(b \mid c) = P(b)$.
- (c) If $P(a \mid b) = P(a)$, then $P(a \mid b, c) = P(a \mid c)$.

Exercise 5 Driving styles (AIMA ch. 13)

A self-driving car shares the highway with another car, whose driving style it does not know. It models three styles, cautious (a), normal (b) and aggressive (c), equally likely a priori, under which the other car changes lanes during a given minute with probabilities 0.2, 0.6 and 0.8, respectively. The self-driving car observes the other car for three minutes, recording whether it changes lanes (l) or keeps its lane (k) during each minute, as X_1 , X_2 and X_3 . Let Y be the style of the other car.

- (a) Write the joint distribution $\mathbf{P}(Y, X_1, X_2, X_3)$ of this model as a product of smaller tables, and give these tables.

- (b) Determine the most likely driving style if the other car changed lanes during the first two minutes and kept its lane during the third.

QUIZ

1. The Bayes' rule states that ...

- ☐ $P(x | y) = \frac{P(y | x)P(y)}{P(x)}.$
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- ☐ $P(x) = \frac{P(x | y)P(y)}{P(x)}.$

2. Which of the following is true?

- ☐ $P(x, y | z) = \frac{P(x | y, z)}{P(y, z | x)}.$
- ☐ $P(x, y, z) = P(x)P(y)P(z).$
- ☐ $P(x, y | z) = P(y | x, z)P(x | z).$
- ☐ $P(x | y, z) = \frac{P(x, y, z)}{P(y)P(z)}.$

3. The naive Bayes model assumes ...

- ☐ conditional independence between the effects, given the cause.
- ☐ independence between the effects.
- ☐ independence between the effects and the cause.
- ☐ conditional independence between each effect and the cause, given the other effects.

4. The Bayes' rule can be read as "The posterior is equal to ...

- ☐ the product of the likelihood and the prior".
- ☐ the ratio between the posterior distribution and the evidence, times the prior".
- ☐ the likelihood to prior ratio multiplied by the evidence".
- ☐ the likelihood to evidence ratio multiplied by the prior".